Algorithmic Pseudocode Sample 4

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

```
for t ∈ {−1, . . . , −T^traceback} do                          ▷ Retrieve the values of P_{u,t}
    P_{u,t} ← Power.GetValue(t)
end for
for t ∈ {−1, . . . , −T^traceback} do                  ▷ Initial conditions on the state variables
    if P_{u,t} > MinimumPower.GetValue(t) then
        S_{u,t}^{OFF} ← 0
        S_{u,t}^{STOP} ← 0
        S_{u,t}^{START} ← 0
        if P_{u,t} < P_{u,t−1} then    ▷ Exact initialization required only if the FLAT state is defined.
            S_{t−1}^{ON_UP} ← 0
            S_{t−1}^{ON_DOWN} ← 1
            S_{t−1}^{ON_FLAT} ← 0
        else if P_{u,t} > P_{u,t−1} then
            S_{t−1}^{ON_UP} ← 1
            S_{t−1}^{ON_DOWN} ← 0
            S_{t−1}^{ON_FLAT} ← 0
        else if P_{u,t} = P_{u,t−1} then
            S_{t−1}^{ON_UP} ← 0
            S_{t−1}^{ON_DOWN} ← 0
            S_{t−1}^{ON_FLAT} ← 1
        end if
    else if P_{u,t} > 0 then                          ▷ Reconstruct the startups and shutdowns
        if P_{u,t} < P_{u,t−1} then
            S_t^{STOP} ← 1
            S_t^{START} ← 0
        else
            S_t^{STOP} ← 0
            S_t^{START} ← 1
        end if
    else                                                  ▷ Final possibility: the unit is OFF.
        S_{u,t}^{OFF} ← 1
        S_{u,t}^{STOP} ← 0
        S_{u,t}^{START} ← 0
        S_{t−1}^{ON_UP} ← 0
        S_{t−1}^{ON_DOWN} ← 0
        S_{t−1}^{ON_FLAT} ← 0
    end if
end for
```